RiNEm 2012
Riunione Nazionale di Elettromagnetismo
Roma, 10-14 Settembre 2012

Harmonic Balance Finite Element Analysis of Third Order Intermodulation Products in Ferrite Devices

***Laurent Ntibarikure*, Giuseppe Pelosi, Stefano Selleri**

Department of Electronics and Telecommunications (DET), University of Florence
Via di Santa Marta 3 - 50139 - Florence, Italy

Outline

- Motivation
- Formulation
  - Harmonic Balance Finite Elements for nonlinear microwave devices
- Numerical experiment
  - *H*-plane nonlinear analysis of rectangular waveguide ferrite circulator
- Conclusions

MOTIVATION: THE PROBLEM

- When high power densities are involved within microwave devices, materials exhibit a significant nonlinear behavior:

$$\mathbf{D} = [\bar{\bar{\varepsilon}}(\mathbf{E},\mathbf{H})]\mathbf{E}$$
$$\mathbf{B} = [\bar{\bar{\mu}}(\mathbf{E},\mathbf{H})]\mathbf{H}$$

- If steady state fields are aimed at for a finite set of frequencies, a frequency domain analysis might be preferable to a time domain analysis

- Frequency domain methods for solving nonlinear PDEs require an iterative solution and the inclusion of higher order harmonics to improve accuracy

MOTIVATION: PROPOSED METHODOLOGY

Harmonic Balance Finite Elements:

- A multiharmonic finite elements formulation, the Harmonic Balance Finite Elements (HBFE), allows for the inclusion of higher order harmonics which are coupled by means of material characteristics
- The HBFE approach allows for passive intermodulation analysis (3^{rd} order intermodulation products)

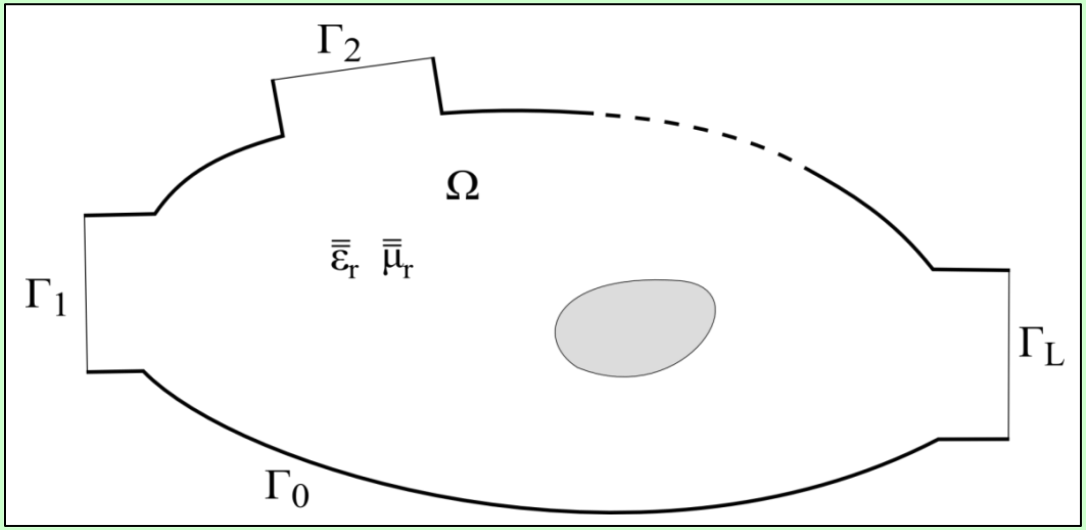
FORMULATION: CONVENTIONAL FINITE ELEMENTS (SINGLE HARMONIC)

- We wish to solve the vector Helmholtz equation

$$\nabla \times \left(\overline{\overline{\mu}}_r^{-1} (\mathbf{E}(\mathbf{r}),\mathbf{r}) \nabla \times \mathbf{E}(\mathbf{r}) \right) - k_0^2 \overline{\overline{\varepsilon}}_r (\mathbf{E}(\mathbf{r}),\mathbf{r}) \mathbf{E}(\mathbf{r}) = 0 \qquad \text{in } \Omega$$

with

- perfect electric conditions on Γ_0
- modal port expansions on Γ_l, l=1…L.



FORMULATION: CONVENTIONAL FINITE ELEMENTS (SINGLE HARMONIC)

- The electric field is expanded in terms of basis functions:

$$\mathbf{E}(\mathbf{r}) = \sum_j e_j \, \mathbf{W}_j(\mathbf{r})$$

- Galerkin projection leads to a linear system of equations:

$$\int_\Omega \mathbf{W}_i(\mathbf{r}) \cdot \left(\nabla \times \left(\overline{\overline{\mu}}_r^{-1} (\mathbf{E}(\mathbf{r}),\mathbf{r}) \nabla \times \mathbf{E}(\mathbf{r}) \right) - k_0^2 \overline{\overline{\varepsilon}}_r (\mathbf{E}(\mathbf{r}),\mathbf{r}) \mathbf{E}(\mathbf{r}) \right) d\Omega = 0$$

$$\Downarrow$$

$$[a_{mn}][E_n] = [b_m]$$

RiNEm 2012
Riunione Nazionale di Elettromagnetismo
Roma, 10-14 Settembre 2012

FORMULATION: HARMONIC BALANCE FE (MULTIPLE HARMONICS)

❖ The electric field is expressed as a sum of harmonic functions:

$$\hat{\mathbf{E}}(\mathbf{r},t) = \sum_{p=1}^{P} \left(\mathbf{E}_p^{(s)}(\mathbf{r}) \sin(\omega_p t) + \mathbf{E}_p^{(c)}(\mathbf{r}) \cos(\omega_p t) \right)$$

where

$$\begin{cases} \mathbf{E}_p^{(s)}(\mathbf{r}) = \sum_j e_{j,p}^{(s)} \mathbf{W}_j(\mathbf{r}) \\ \mathbf{E}_p^{(c)}(\mathbf{r}) = \sum_j e_{j,p}^{(c)} \mathbf{W}_j(\mathbf{r}) \end{cases}$$

S. Yamada, K. Bessho, and J. Lu, "Harmonic balance finite element method applied to nonlinear AC magnetic analysis," *IEEE Trans. Magn.*, **MAG-25**, no. 4, pp. 2971–2973, Jul. 1989

D. M. Copeland and U. Langer, "Domain decomposition solvers for nonlinear multiharmonic finite element equations," *J. Numer. Math.*, vol. 18, no. 3, pp. 157–175, Oct. 2010

RiNEm 2012
Riunione Nazionale di Elettromagnetismo
Roma, 10-14 Settembre 2012

FORMULATION: HARMONIC BALANCE FE (MULTIPLE HARMONICS)

- Materials permittivity or permeability is also expressed in terms of harmonic functions:

$$\overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r},t) = \overline{\overline{\xi}}_{r_0} + \sum_{g=1}^{G}\left(\overline{\overline{\xi}}_{r_g}^{(s)}\sin(\omega_g t) + \overline{\overline{\xi}}_{r_g}^{(c)}\cos(\omega_g t)\right)$$

$$\overline{\overline{\xi}}_r \equiv \left\{\overline{\overline{\mu}}_r, \overline{\overline{\varepsilon}}_r\right\}$$

with

$$\begin{cases} \overline{\overline{\xi}}_{r_0} = \dfrac{\omega}{2\pi}\displaystyle\int_0^{\frac{2\pi}{\omega}} \overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\,dt \\ \overline{\overline{\xi}}_{r_g}^{(s)} = \dfrac{\omega}{\pi}\displaystyle\int_0^{\frac{2\pi}{\omega}} \overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\sin(\omega_g t)\,dt \\ \overline{\overline{\xi}}_{r_g}^{(c)} = \dfrac{\omega}{\pi}\displaystyle\int_0^{\frac{2\pi}{\omega}} \overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\cos(\omega_g t)\,dt \end{cases}$$

FORMULATION: HARMONIC BALANCE FE (MULTIPLE HARMONICS)

- With further Galerkin projection of the vector Helmholtz equation with the chosen harmonic functions ($k_0=\omega_p/c_0$), a large scale HBFE system is obtained

$$\left[a_{mn}\right]\left[E_n\right]=\left[b_m\right]$$

$$\Downarrow$$

$$a_{mn}=\begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \cdots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \cdots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \cdots & a_{mn}^{(PcPc)} \end{bmatrix} \quad E_n=\begin{bmatrix} E_n^{(1s)} \\ E_n^{(1c)} \\ \vdots \\ E_n^{(Ps)} \end{bmatrix} \quad b_m=\begin{bmatrix} b_m^{(1s)} \\ b_m^{(1c)} \\ \vdots \\ b_m^{(Pc)} \end{bmatrix}$$

FORMULATION: HARMONIC BALANCE FE (MULTIPLE HARMONICS)

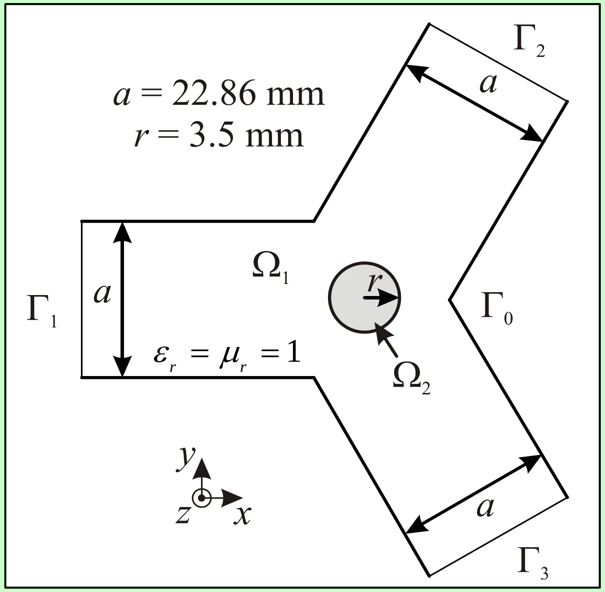
- The nonlinear solution is handled via a relaxed iteration, starting from a null solution vector to evaluate material properties

$$[A_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}][E]^i = [b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}], \qquad i \geq 2,\ \gamma \in (0,1]$$

- The convergence criterion is chosen on the basis of consecutive solutions:

$$\frac{\left\|[E]^i - [E]^{i-1}\right\|}{\left\|[E]^i\right\|} \leq \tau$$

NUMERICAL EXPERIMENT: LINEAR MICROWAVE FERRITE CIRCULATOR



- For *H*-plane uniformity

$$\mathbf{E}(\mathbf{r}) = E_z\,\hat{\mathbf{z}} = \sum_{n=1}^{N} E_n \varphi_n\,\hat{\mathbf{z}}$$

Thus, vector formulation can be treated as scalar:

$$\int_\Omega \mathbf{W}_i(\mathbf{r}) \cdot \left(\nabla \times \left(\overline{\overline{\mu}}_r^{\,-1}(\mathbf{E}(\mathbf{r}),\mathbf{r}) \nabla \times \mathbf{E}(\mathbf{r}) \right) - k_0^2\, \overline{\overline{\varepsilon}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) \mathbf{E}(\mathbf{r}) \right) d\Omega = 0$$

$$\Downarrow$$

$$\int_\Omega \nabla_t \varphi_m \cdot \overline{\overline{\mu}}_r^{\,-1} \nabla_t E_z\, d\Omega - k_0^2 \varepsilon_r \int_\Omega \varphi_m \cdot E_z\, d\Omega - \sum_{k=1}^{3} \int_{\Gamma_k} \varphi_m \frac{\partial E_z}{\partial n^{(k)}}\, d\Gamma = 0$$

NUMERICAL EXPERIMENT: LINEAR MICROWAVE FERRITE CIRCULATOR

$$\overline{\overline{\mu_r}}(\mathbf{r}) = \begin{bmatrix} \mu_r & j\kappa_r & 0 \\ -j\kappa_r & \mu_r & 0 \\ 0 & 0 & 1 \end{bmatrix}, \ \mathbf{r} \in \Omega_2$$

$$\mu_r = 1 + \frac{\omega_0 \omega_m}{{\omega_0}^2 - \omega^2}$$

$$\kappa_r = \frac{\omega \omega_m}{{\omega_0}^2 - \omega^2}$$

$$\omega_0 = \gamma H_0, \ \ \omega_m = \gamma M_s$$

M. Koshiba and M. Suzuki, "Finite-Element Analysis of *H*-Plane Waveguide Junction with Arbitrarily Shaped Ferrite Post," *IEEE Trans. Microw. Theory Tech.*, **MTT-34**, no. 1, pp. 103–109, Jan. 1986.

NUMERICAL EXPERIMENT: LINEAR MICROWAVE FERRITE CIRCULATOR

- The internal DC magnetic field H_0 , out-of-plane directed, is considered uniform within the TT1-109 ferrite post

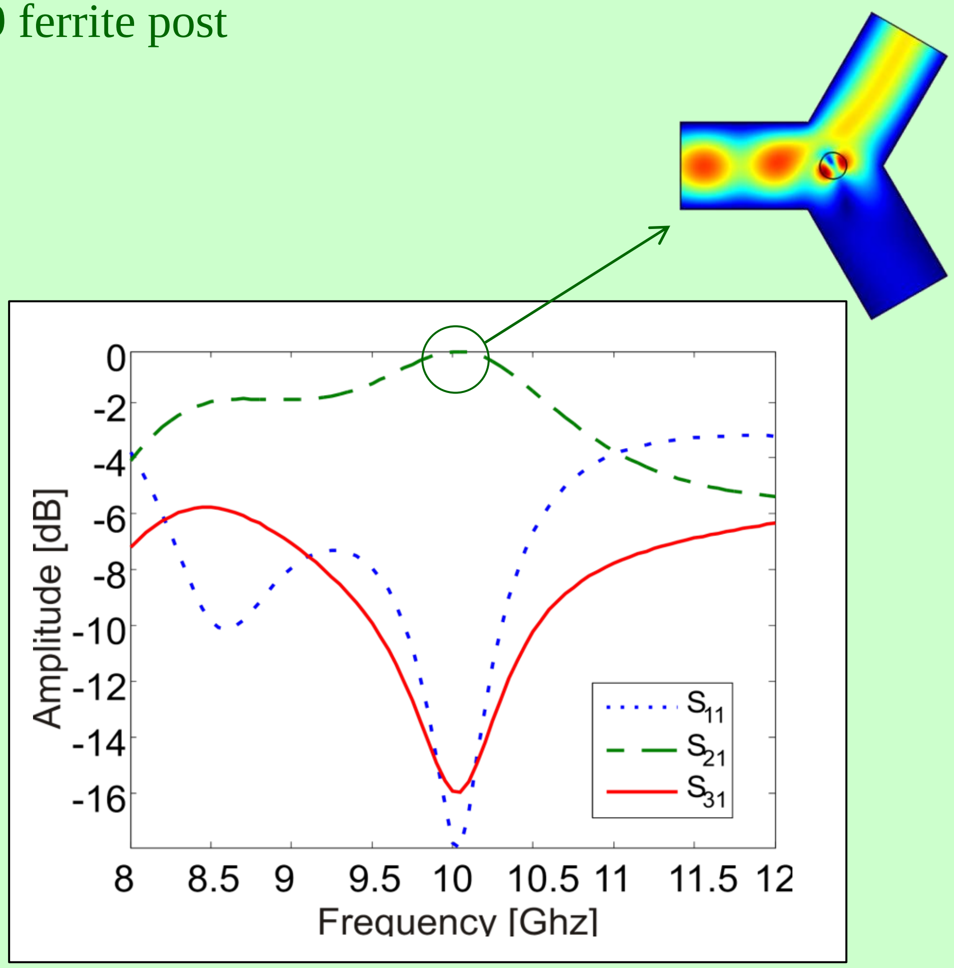
TT1-109 Ferrite Post :

$$\gamma = 1.76 \cdot 10^7 \left[C/kg \right]$$

$$H_o = 200 \left[G \right]$$

$$M_s = 1317 \left[\mathrm{Oe} \right]$$

$$\varepsilon_r(\mathbf{r}) = 11.7, \ \mathbf{r} \in \Omega_2$$



NUMERICAL EXPERIMENT: NONLINEAR MICROWAVE FERRITE CIRCULATOR

- To derive a nonlinear permeability of the ferrite at microwave frequencies, the total magnetic induction and field are given by:

$$\mathbf{H} = H_0 \hat{z} + \mathbf{h}_{\mathrm{AC}}, \ \mathbf{M} = M_s \hat{z} + \mathbf{m}_{\mathrm{AC}}$$

- The macroscopic equation of motion is

$$\frac{\partial \mathbf{M}}{\partial t} = \mu_0 \gamma \left(\mathbf{H} \times \mathbf{M} \right)$$

and expressing equation for each component of the magnetization in frequency domain:

$$\begin{cases} j\omega m_x = \mu_0 \gamma \left(h_x \left(M_s + m_z \right) - m_y \left(H_i + h_z \right) \right) \\ j\omega m_y = \mu_0 \gamma \left(h_x \left(M_s + m_z \right) + m_x \left(H_i + h_z \right) \right) \\ j\omega m_z = \mu_0 \gamma \left(h_x m_y - m_x h_y \right) \end{cases}$$

NUMERICAL EXPERIMENT: NONLINEAR MICROWAVE FERRITE CIRCULATOR

- By solving the previous system of equations, the following nonlinear relative permeability tensor is obtained:

$$\overline{\overline{\mu_r}}(\mathbf{H}) = \begin{bmatrix} \mu_r(\mathbf{H}) & j\kappa_r(\mathbf{H}) & 0 \\ -j\kappa_r(\mathbf{H}) & \mu_r(\mathbf{H}) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mu_r = 1 + \frac{\omega_0 \omega_m}{\omega_0^{\,2} + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2} \approx \overline{\mu_r} + \alpha\left(h_x^2 + h_y^2\right)$$

$$\kappa_r = \frac{\omega \omega_m}{\omega_0^{\,2} + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2} \approx \overline{\kappa_r} + \alpha\left(h_x^2 + h_y^2\right)$$

NUMERICAL EXPERIMENT: NONLINEAR MICROWAVE FERRITE CIRCULATOR

- Two signals at ω_1 and ω_2 impinging on the device are considered. The out-of-plane electric field is hence expressed in terms of harmonic functions as

$$
\begin{aligned}
\hat{E}_z(\mathbf{r}) = {} & E_{z_1}^{(s)} \sin(\omega_1 t) + E_{z_2}^{(s)} \sin(\omega_2 t) + E_{z_3}^{(s)} \sin((2\omega_1 - \omega_2)t) + E_{z_4}^{(s)} \sin((2\omega_2 - \omega_1)t) + \\
& E_{z_5}^{(s)} \sin((2\omega_1 + \omega_2)t) + E_{z_6}^{(s)} \sin((2\omega_2 + \omega_1)t) + E_{z_7}^{(s)} \sin(3\omega_1 t) + E_{z_8}^{(s)} \sin(3\omega_2 t) + \dots \\
& E_{z_1}^{(c)} \cos(\omega_1 t) + E_{z_2}^{(c)} \cos(\omega_2 t) + E_{z_3}^{(c)} \cos((2\omega_1 - \omega_2)t) + E_{z_4}^{(c)} \cos((2\omega_2 - \omega_1)t) + \\
& E_{z_5}^{(c)} \cos((2\omega_1 + \omega_2)t) + E_{z_6}^{(c)} \cos((2\omega_2 + \omega_1)t) + E_{z_7}^{(c)} \cos(3\omega_1 t) + E_{z_8}^{(c)} \cos(3\omega_2 t) + \dots
\end{aligned}
$$

and so will be the magnetic in-plane components.

NUMERICAL EXPERIMENT: NONLINEAR MICROWAVE FERRITE CIRCULATOR

- Ferrite permeability is also expressed in terms of harmonic functions, with coefficients given by, for example for the diagonal ones of the permeability tensor:

$$\hat{\mu}_r = \mu_{r_0} + \sum_{m+n>0 \text{ even}} \mu_{r^+_{m,n}} \cos(m\omega_1 t + n\omega_2 t) + \mu_{r^-_{m,n}} \cos(m\omega_1 t - n\omega_2 t)$$

with

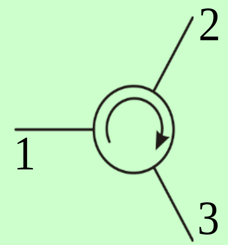
$$\begin{cases} \mu_{r_0} = \dfrac{\gcd(\omega_1,\omega_2)}{2\pi} \displaystyle\int_0^{\frac{2\pi}{\gcd(\omega_1,\omega_2)}} \mu_r(\mathbf{H})\,dt \\ \mu^{(s)}_{r^{\pm}_{m,n}} = \dfrac{\gcd(\omega_1,\omega_2)}{\pi} \displaystyle\int_0^{\frac{2\pi}{\gcd(\omega_1,\omega_2)}} \mu_r(\mathbf{H})\sin(m\omega_1 t \pm n\omega_2 t)\,dt \\ \mu^{(c)}_{r^{\pm}_{m,n}} = \dfrac{\gcd(\omega_1,\omega_2)}{\pi} \displaystyle\int_0^{\frac{2\pi}{\gcd(\omega_1,\omega_2)}} \mu_r(\mathbf{H})\cos(m\omega_1 t \pm n\omega_2 t)\,dt \end{cases}$$

and

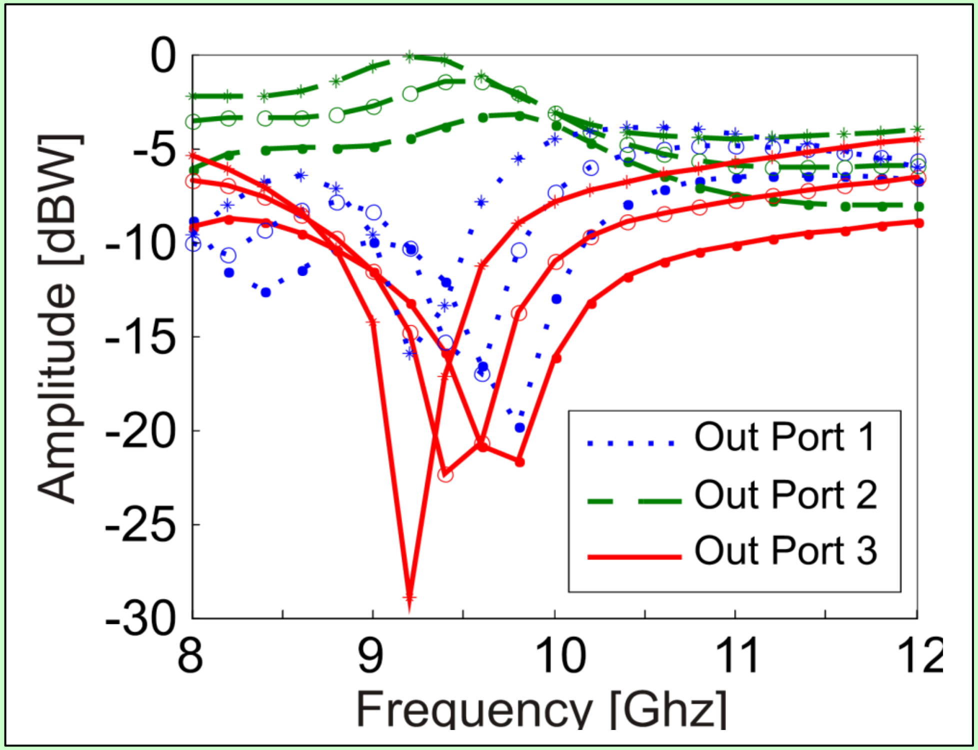
$$\mu_r(\mathbf{H}) = \overline{\mu_r} + \alpha\left(h_x^2 + h_y^2\right)$$

NUMERICAL EXPERIMENT: NONLINEAR MICROWAVE FERRITE CIRCULATOR

$$P_s \ @\ f_1 \in [8,12]\ \text{GHz}$$
$$P_i \ @\ f_2 = 7.8\ \text{GHz}$$



$$\alpha = 2 \cdot 10^{-7}\ \text{m}^2/\text{A}^2$$



• $\begin{cases} P_s = 0.5\,\text{W} \\ P_i = 50\,\text{W} \end{cases}$

○ $\begin{cases} P_s = 0.75\,\text{W} \\ P_i = 75\,\text{W} \end{cases}$

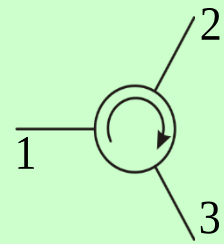
$*$ $\begin{cases} P_s = 1\,\text{W} \\ P_i = 100\,\text{W} \end{cases}$

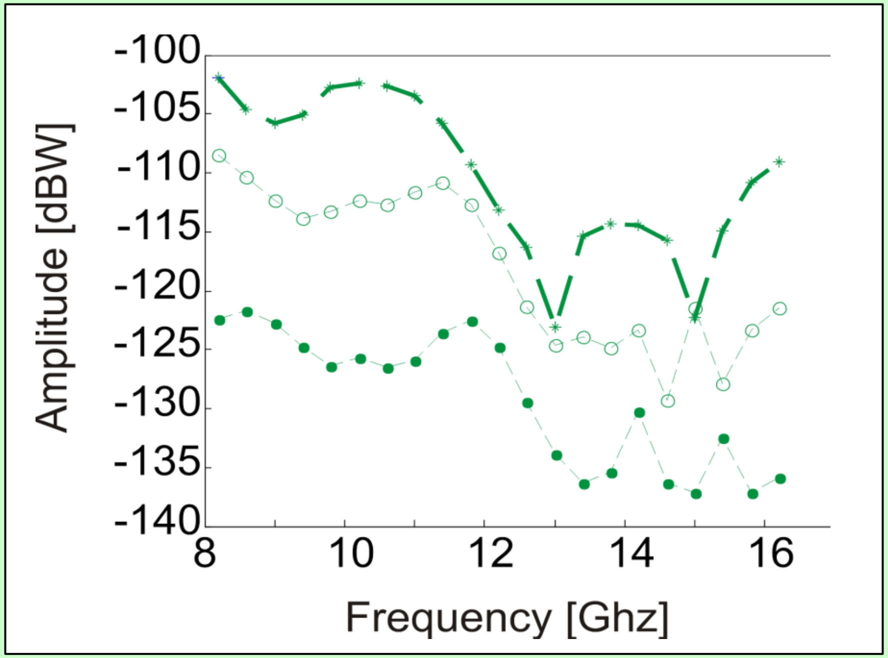
NUMERICAL EXPERIMENT: NONLINEAR MICROWAVE FERRITE CIRCULATOR

$$P^{\mathrm{IM3}} @\, 2f_1 - f_2$$

$$P_s \;@\; f_1 \in [8,12]\,\mathrm{GHz}$$

$$P_i \,@\; f_2 = 7.8\,\mathrm{GHz}$$





• $\begin{cases} P_s = 0.5\,\mathrm{W} \\ P_i = 50\mathrm{W} \end{cases}$

○ $\begin{cases} P_s = 0.75\,\mathrm{W} \\ P_i = 75\mathrm{W} \end{cases}$

* $\begin{cases} P_s = 1\,\mathrm{W} \\ P_i = 100\mathrm{W} \end{cases}$

H. How, C. Vittoria and R. Schmidt, "Nonlinear intermodulation coupling in ferrite circulator junctions," *IEEE Trans. Microw. Theory Tech.*, MTT-45, no. 3, pp. 245–252, Feb. 1997.

CONCLUSIONS

- A Harmonic Balance Finite Element technique has been presented for solving the wave equation in presence of nonlinear media
- Third order intermodulation products within a nonlinear ferrite circulator have been evaluated, in good agreement with the reported litterature